MERLIN: Derivations and Proofs for the Bilateral Controller

Companion document to the MERLIN paper and repository

Abstract

This document states, for every equation of the revised MERLIN controller, the assumptions under which it holds, the result, and a proof. Each statement that can be checked numerically is tied to a unit test in `ros_ws/src/controller/test/test_derivations.py`. Section 10 lists what is *not* proven. It also explains, equation by equation, how the revised formulas differ from the equations of the submitted draft.

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1 Scope, notation and standing assumptions

The teleoperation pair is a ReactorX 200 master and a ViperX 300s slave with the forearm-roll joint locked at zero. Each arm has $n = 5$ active joints $q = [q_{\text{waist}}, q_{\text{shoulder}}, q_{\text{elbow}}, q_{\text{wrist angle}}, q_{\text{wrist rotate}}]^\top$. The controller runs at 50 Hz, $\Delta t = 0.02\,\text{s}$. Subscripts m and s denote master and slave. For a vector x, $\text{diag}(x)$ is the diagonal matrix, $\odot$ the elementwise product, $\text{sat}_a(x) = \text{clip}(x, -a, a)$ and

$\mathrm{dz}_\delta(x) = \mathrm{sign}(x)\max(|x| - \delta, 0)$ act elementwise. $\mathcal{F}_\alpha\{x\}[k] = \alpha x[k] + (1-\alpha)\mathcal{F}_\alpha\{x\}[k-1]$ is a first-order IIR filter.

Assumption 1 (rigid-body model)**.** For each arm, $M(q) = M(q)^\top \succ 0$ is the joint-space inertia matrix, $C(q, \dot{q})$ is the Coriolis matrix built from the Christoffel symbols, and $G(q)$ the gravity torque, so that

$$M(q)\ddot{q} + C(q, \dot{q})\dot{q} + G(q) = \tau_m + \tau_{ext}. \tag{1}$$

τ_m is the motor torque and τ_{ext} the *external torque*: everything not in the model, i.e. contact and operator forces, joint friction and model error. The Christoffel construction gives $\dot{M} = C + C^\top$, hence $\dot{q}^\top(\dot{M} - 2C)\dot{q} = 0$.

Assumption 2 (actuation and sensing)**.** Motor torque is $\tau_m = n_\mu \odot k_t \odot 10^{-3} I$, with I the present current in mA (as published by the Interbotix `xs_sdk`), k_t the torque constant in N m/A, and n_μ the number of motors per joint (two for the shoulder of the RX200 and for the shoulder and elbow of the VX300s). The current loop is much faster than 50 Hz. Joint positions are measured; $\dot{q}$ is obtained from them.

Assumption 3 (slave position loop)**.** Each slave joint is driven by a Dynamixel position loop, i.e. it accepts a position command u, not a torque. Seen from u, joint i behaves as a decoupled second-order system with a lumped, slowly varying acceleration disturbance c_i (gravity sag, friction):

$$\ddot{q}_i = \omega_i^2(u_i - q_i) - 2\zeta_i\omega_i\dot{q}_i + c_i. \tag{2}$$

ω_i, ζ_i are identified per joint (`tools/identify_position_loop.py`). This is a modelling assumption, validated in simulation only.

Sign convention. τ_{ext} is the torque the environment applies *on* the joint. The torque the controller renders on the operator points in the same direction as the environment's torque on the slave (the operator is pushed back).

2 External-torque estimation

2.1 The static residual and why it fails during motion

The draft estimates $\hat{\tau}_{ext} = \tau_m - (C - G)$; with Assumption 1 the correct static residual is the following.

Lemma 1 (static residual)**.** *Define* $\hat{\tau}^{static} := G(q) + C(q, \dot{q})\dot{q} - \tau_m$. *Then* $\hat{\tau}^{static} = \tau_{ext} - M(q)\ddot{q}$.

Proof. Solve (1) for τ_{ext}: $\tau_{ext} = M\ddot{q} + C\dot{q} + G - \tau_m$, and subtract $M\ddot{q}$. □

So the residual is correct only when $M\ddot{q}$ is negligible. On a position-controlled arm the position loop accelerates the links, and $M\ddot{q}$ appears as a false contact torque. In the MuJoCo benchmark the shoulder inertia is $M_{22} \approx 0.23$–$0.30\,\mathrm{kg\,m^2}$ and the free-motion false torque grows with speed as 0.25, 0.48, 1.08 N m at $1\times, 2\times, 3\times$ the motion speed; its correlation with $-M\ddot{q}$ is 0.67, 0.81, 0.92. At low speed friction, mass mismatch and current noise dominate; at high speed $M\ddot{q}$ does.

Numerical check: `test_static_residual_equals_tau_ext_minus_Mqdd, test_inverse_dynamics_decomposition`.

2.2 Generalised-momentum observer

Define the generalised momentum $p := M(q)\dot{q}$ and $\beta(q, \dot{q}) := G(q) - C(q, \dot{q})^\top\dot{q}$.

Lemma 2 (momentum balance)**.** *Under Assumption 1,* $\dot{p} = \tau_m + \tau_{ext} - \beta(q, \dot{q})$.

Proof. $\dot{p} = \dot{M}\dot{q} + M\ddot{q}$. Substituting (1), $M\ddot{q} = \tau_m + \tau_{ext} - C\dot{q} - G$, and $\dot{M} = C + C^\top$, $\dot{p} = (C + C^\top)\dot{q} + \tau_m + \tau_{ext} - C\dot{q} - G = \tau_m + \tau_{ext} + C^\top\dot{q} - G$. □

Theorem 3 (observer)**.** *Let $K_O = \mathrm{diag}(k_1, \ldots, k_n)$, $k_i > 0$, and define*

$$r(t) = K_O\Big(p(t) - p(0) - \int_0^t \big(\tau_m + C^\top\dot{q} - G + r\big)\,ds\Big). \tag{3}$$

Then $r(0) = 0$ and $\dot{r} = K_O(\tau_{ext} - r)$. In particular r_i is a first-order low-pass filter of $\tau_{ext,i}$ with cutoff k_i rad/s, $r_i(t) = \int_0^t e^{-k_i(t-s)}k_i\,\tau_{ext,i}(s)\,ds$, and $r \to \tau_{ext}$ for a constant τ_{ext}. No acceleration is needed.

Proof. $r(0) = 0$ by construction. Differentiating (3), $\dot{r} = K_O(\dot{p} - \tau_m - C^\top\dot{q} + G - r)$. By Lemma 2, $\dot{p} - \tau_m - C^\top\dot{q} + G = \tau_{ext}$. The solution of the scalar linear ODE $\dot{r}_i = k_i(\tau_{ext,i} - r_i)$, $r_i(0) = 0$ is the stated convolution; for τ_{ext} constant it equals $\tau_{ext,i}(1 - e^{-k_i t})$. □

Numerical check: `test_Mdot_equals_C_plus_Ct, test_observer_converges_to_constant_external_torque`.

Remark 4 (bias from model error)*.* If the implementation uses imperfect $\tilde{M}, \tilde{G}, \tilde{C}, \tilde{k}_t$, repeating the proof gives $\dot{\tilde{r}} = K_O(\tau_{ext} + \delta - \tilde{r})$ with

$$\delta = (\tau_m - \tilde{\tau}_m) + (\tilde{\beta} - \beta) + \Delta\dot{M}\,\dot{q} + \Delta M\,\ddot{q}, \qquad \Delta M = \tilde{M} - M.$$

So the estimate is the low-passed external torque *plus* a bias from the torque constants, from gravity and Coriolis errors, and from the inertia error times the acceleration (which is small: $\Delta M\ddot{q}$, not $M\ddot{q}$). In the simulation, link masses are 8% higher than the controller's model, so the gravity term alone gives $\delta \approx 0.08\,G \approx 0.2\,\mathrm{N\,m}$ at the shoulder and elbow ($G = [-2.3, -3.0]\,\mathrm{N\,m}$ at the reference pose), the same order as the 0.25–0.3 N m RMS error measured in the estimation study. On the real arms, k_t and the link masses must be calibrated with known weights (`docs/HARDWARE_VALIDATION.md`).

Proposition 5 (discrete observer)**.** *Implement (3) at sample time Δt with the rectangle rule: $S[k] = S[k-1] + \Delta t\,(\tau_m[k] - \beta[k] + r[k-1])$, $r[k] = K_O(p[k] - p[0] - S[k])$. If $p[k] - p[k-1] = \Delta t(\tau_m[k] - \beta[k] + \tau_{ext}[k]) + \varepsilon[k]$, with $\varepsilon[k]$ the quadrature error of the integral of $\dot{p}$, then*

$$r[k] = (1 - K_O\Delta t)\,r[k-1] + K_O\Delta t\,\tau_{ext}[k] + K_O\varepsilon[k]. \tag{4}$$

It is a stable first-order IIR filter iff $0 < k_i\Delta t < 2$, and monotone iff $k_i\Delta t < 1$. With $K_O = 15$ rad/s, $\Delta t = 0.02$ s: $a = K_O\Delta t = 0.3$, equivalent time constant $-\Delta t/\ln(1-a) = 56$ ms (the continuous value is $1/K_O = 67$ ms).

Proof. Subtract the update for $r[k-1]$ from the one for $r[k]$: $r[k] - r[k-1] = K_O\big((p[k] - p[k-1]) - \Delta t(\tau_m[k] - \beta[k] + r[k-1])\big)$. Insert the balance for $p[k] - p[k-1]$ to get $r[k] - r[k-1] = K_O\Delta t(\tau_{ext}[k] - r[k-1]) + K_O\varepsilon[k]$, which is (4). The pole is $1 - k_i\Delta t$, inside the unit circle iff $0 < k_i\Delta t < 2$. □

Numerical check: `test_discrete_observer_recursion_is_exact_iir (the recursion holds to` 10^{-10} `when` $\varepsilon = 0$`)`.

Remark 6 (unfiltered inputs; empirical)*.* The implementation feeds the observer the unfiltered velocity and torque. The velocity IIR filter of the draft ($\alpha_v = 0.3$) has a mean delay of $(1-\alpha_v)/\alpha_v = 2.3$ samples $= 47$ ms, comparable with the observer time constant. Feeding the filtered velocity delayed $p = M\dot{q}$ relative to $\int\tau$ and, in the simulation, produced a master–slave limit cycle (free-motion RMSE 0.113 rad against 0.016 rad). This is an empirical observation, not a theorem.

3 Slave admittance: exact discretisation

For each slave joint, the compliance offset q_e obeys the desired impedance

$$M_d\ddot{q}_e + D_d\dot{q}_e + K_d q_e = u, \qquad u = \mathrm{dz}_\delta(\hat{\tau}_{ext,s}), \qquad q_{s,\mathrm{ref}} = Sq_m + o + q_e, \tag{5}$$

with design parameters $M_d, D_d, K_d > 0$ (not the physical inertia). With $x = [q_e, \dot{q}_e]^\top$, $A = \begin{bmatrix} 0 & 1 \\ -K_d/M_d & -D_d/M_d \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ 1/M_d \end{bmatrix}$ we have $\dot{x} = Ax + Bu$.

Theorem 7 (exact zero-order-hold discretisation)**.** *If u is constant on $[k\Delta t, (k+1)\Delta t)$, then $x[k+1] = A_d x[k] + B_d u[k]$ exactly, with $A_d = e^{A\Delta t}$, $B_d = \int_0^{\Delta t} e^{As} B\, ds$, and*

$$\exp\Big(\Delta t \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}\Big) = \begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix}. \tag{6}$$

Proof. Variation of constants gives $x(t_{k+1}) = e^{A\Delta t}x(t_k) + \int_0^{\Delta t} e^{A(\Delta t - s)} B\, u[k]\, ds$, and the substitution $s \mapsto \Delta t - s$ turns the integral into $B_d u[k]$. For (6), let $E(t) = \exp(t\bar{A})$ with $\bar{A} = \begin{bmatrix} A & B \\ 0 & 0 \end{bmatrix}$ and write $E(t) = \begin{bmatrix} E_{11} & F \\ 0 & 1 \end{bmatrix}$. From $\dot{E} = \bar{A}E$: $\dot{E}_{11} = AE_{11}$, $E_{11}(0) = I$, so $E_{11} = e^{At}$; and $\dot{F} = AF + B$, $F(0) = 0$, so $F = \int_0^t e^{A(t-s)} B\, ds = \int_0^t e^{As} B\, ds$ (Van Loan's formula [1]). □

Numerical check: `test_zoh_matches_fine_integration (agrees with a` 10^{-12}`-tolerance ODE solve to` 10^{-10}`)`.

Proposition 8 (stability; failure of explicit Euler)**.** *(i) For $M_d, D_d, K_d > 0$ every eigenvalue λ of A has $\mathrm{Re}\,\lambda < 0$, so the eigenvalues $e^{\lambda\Delta t}$ of A_d satisfy $|e^{\lambda\Delta t}| = e^{\mathrm{Re}\,\lambda\,\Delta t} < 1$ for* every *$\Delta t > 0$. (ii) The static deflection is $q_e = u/K_d$. (iii) Explicit Euler, $x[k+1] = (I + \Delta t A)x[k] + \Delta t B u[k]$, is stable iff $|1 + \lambda\Delta t| < 1$ for both eigenvalues.*

Proof. (i) The eigenvalues are the roots of $M_d s^2 + D_d s + K_d = 0$; a real quadratic with all coefficients of the same sign has both roots in the open left half plane (Routh–Hurwitz). (ii) Set $\dot{x} = 0$. (iii) The eigenvalues of $I + \Delta t A$ are $1 + \lambda\Delta t$. □

Remark 9 (worked example: the original wrist). The original code used the robot inertia as M_d: for the wrist rotate, $M_d = 7.29 \times 10^{-4}\,\mathrm{kg\,m^2}$, with $D_d = 0.4$, $K_d = 10$. The roots are $\lambda = -26.3$ and $-522.4\,\mathrm{s}^{-1}$ ($\omega_n = 117\,\mathrm{rad/s}$, $\zeta = 2.34$). At $\Delta t = 20\,\mathrm{ms}$, $|1 + \lambda\Delta t| = 0.47$ and 9.45: Euler diverges, whereas $|e^{\lambda\Delta t}| = 0.59$ and 3×10^{-5}. The original code kept the state bounded only with ad-hoc clamps ($\pm 0.05\,\mathrm{rad}$, $\ddot{q}_e \le 30\,\mathrm{rad/s^2}$), which chatter. The revised design values $M_d = [0.30, 0.40, 0.30, 0.10, 0.04]$, $D_d = [6, 8, 6, 2, 0.8]$, $K_d = [30, 40, 30, 10, 4]$ give $\omega_n = \sqrt{K_d/M_d} \approx 10\,\mathrm{rad/s}$, $\zeta = 1$.

Numerical check: `test_euler_unstable_for_original_wrist_admittance, test_admittance_static_deflection_is_tau_over_K`.

Remark 10. The implementation saturates $|q_e| \le q_{e,\max}$ and zeroes $\dot{q}_e$ when pushing into the limit (anti-windup). The saturated filter is bounded by construction; Theorem 7 and Proposition 8 describe the unsaturated range.

4 Master law

The RX200 master runs in Dynamixel current mode, so the commanded current is applied as torque. Its *measured* current is therefore, up to noise and delay, the *commanded* current.

Lemma 11 (the original law integrates its own command)**.** *Let the measured torque be the command delayed by one sample and IIR-filtered with coefficient α, $\tilde{\tau}[k] = \mathcal{F}_\alpha\{\tau_{cmd}[k-1]\}$. The original law is $\tau_{cmd} = \hat{G} - D_{md}\dot{q} + \hat{\tau}_{ext,m} + \tau_f$ with $\hat{\tau}_{ext,m} = \tilde{\tau} - C - \hat{G}$ (its own estimator). Then $\tau_{cmd}[k] = \tilde{\tau}[k] + b[k]$ with $b = \tau_f - D_{md}\dot{q} - C\dot{q}$, which has a pole at $z = 1$. A constant $b \neq 0$ makes τ_{cmd} ramp by αb per sample until the clamp.*

Proof. Substitute $\hat{\tau}_{ext,m}$: the $\hat{G}$ terms cancel, so $\tau_{cmd}[k] = \tilde{\tau}[k] + b[k]$. With $\tilde{\tau}[k] = \alpha\tau_{cmd}[k-1] + (1-\alpha)\tilde{\tau}[k-1]$ the transfer function from b to τ_{cmd} is $\big(1 - (1-\alpha)z^{-1}\big)/(1 - z^{-1})$: its DC gain of the feedback path is 1, so the pole is at $z = 1$. For constant b: $\tilde{\tau}[k] = \tilde{\tau}[k-1] + \alpha b$ (induction from $\tilde{\tau}[k] = \alpha(\tilde{\tau}[k-1] + b) + (1-\alpha)\tilde{\tau}[k-1]$). □

Numerical check: `test_original_master_law_has_a_pole_at_one`.

Any constant bias in b (a gravity-model error, a sensor offset, the slave's coupling term) is integrated until the torque clamp: this is the "rough or no change" behaviour reported on hardware. The revised law removes $\hat{\tau}_{ext,m}$:

$$\tau_{m,cmd} = \hat{G}_m(q_m) - D_{md}\dot{q}_m + \gamma\,\tau_f, \tag{7}$$

where τ_f depends on slave data and on $q_m - q_s$ only, and $\gamma \in [0, 1]$ is the tank gate of Section 7. There is no path from the master's measured current to its command, so the pole of Lemma 11 does not arise.

Proposition 12 (passivity of the master with the revised law)**.** *Let $\hat{G}_m = G_m$ and let the operator apply $\tau_h = J_m^\top F_h$. With $V = \frac{1}{2}\dot{q}_m^\top M_m \dot{q}_m$,*

$$\dot{V} = \dot{q}_m^\top \tau_h - \dot{q}_m^\top D_{md}\dot{q}_m + \gamma\,\dot{q}_m^\top \tau_f.$$

Proof. $\dot{V} = \dot{q}^\top M\ddot{q} + \frac{1}{2}\dot{q}^\top \dot{M}\dot{q} = \dot{q}^\top(\tau_{m,cmd} + \tau_h - C\dot{q} - G) + \frac{1}{2}\dot{q}^\top \dot{M}\dot{q}$. By Assumption 1, $\dot{q}^\top(\frac{1}{2}\dot{M} - C)\dot{q} = 0$. Insert (7); the gravity terms cancel. □

With $\tau_f = 0$ the master is output-strictly passive from τ_h to $\dot{q}_m$. The last term, $\gamma\dot{q}_m^\top\tau_f = \gamma P_{out}$, is the power the force channel delivers to the operator. It is exactly the quantity the energy tank of Section 7 bounds. Gravity compensation is approximate on hardware, so the statement holds up to $\dot{q}^\top(\hat{G} - G)$.

5 Force reflection between kinematically different arms

Reflecting joint torque one-to-one renders the wrong Cartesian force when the arms differ. At the reference pose the RX200 reaches 0.388 m and the VX300s 0.530 m (ratio 1.37).

Proposition 13 (damped least-squares wrench)**.** *Let $J_s \in \mathbb{R}^{6\times n}$ be the slave end-effector Jacobian (full column rank $n = 5$) and $\lambda > 0$. The wrench*

$$\hat{F}_s = \arg\min_F \ \|J_s^\top F - \hat{\tau}_{ext,s}\|^2 + \lambda^2\|F\|^2 = J_s\big(J_s^\top J_s + \lambda^2 I\big)^{-1}\hat{\tau}_{ext,s} \tag{8}$$

satisfies $J_s^\top \hat{F}_s = \hat{\tau}_{ext,s} - \lambda^2(J_s^\top J_s + \lambda^2 I)^{-1}\hat{\tau}_{ext,s}$, i.e. $J_s^\top \hat{F}_s \to \hat{\tau}_{ext,s}$ as $\lambda \to 0$.

Proof. Setting the gradient to zero, $J_s(J_s^\top F - \hat{\tau}) + \lambda^2 F = 0$, so $(J_s J_s^\top + \lambda^2 I)F = J_s\hat{\tau}$. The push-through identity $(J_s J_s^\top + \lambda^2 I)^{-1} J_s = J_s(J_s^\top J_s + \lambda^2 I)^{-1}$ gives (8). Then $J_s^\top \hat{F}_s = J_s^\top J_s(J_s^\top J_s + \lambda^2 I)^{-1}\hat{\tau} = \hat{\tau} - \lambda^2(J_s^\top J_s + \lambda^2 I)^{-1}\hat{\tau}$. □

The 6-dimensional wrench is not unique because five joints cannot produce one wrench direction; (8) selects the minimum-norm one. The torque rendered on the master is

$$\tau_f = k_{of}\, J_m^\top S_f\, \hat{F}_s - k_{om} K_{md}(S q_m + o - q_s), \qquad S_f = \mathrm{diag}(s_f I_3,\, s_\mu I_3). \tag{9}$$

(the second term is the position channel; $\hat{\tau}_{ext,s}$ in (8) is first passed through dz_δ).

Proposition 14 (consistency)**.** *If the two arms have the same Jacobian at their configurations, $S_f = I$ and $\lambda \to 0$, then $J_m^\top \hat{F}_s = \hat{\tau}_{ext,s}$: the Jacobian map reduces to the identity map. For two different arms and the same joint torque τ, the identity map yields a master-side force $F_{\mathrm{id}} = (J_m^\top)^+ \tau$ whereas the force that produced τ on the slave is $F_s = (J_s^\top)^+ \tau$; at the reference pose $\|F_{\mathrm{id}}\|/\|F_s\| \approx 1.3$.*

Proof. The first claim is Proposition 13 with $J_m = J_s$. The ratio follows from $\tau = J^\top F$ (virtual work, $\delta W = \tau^\top \delta q = F^\top J \delta q$): a shorter arm needs a larger force for the same joint torque. □

Numerical check: `test_dls_wrench_reproduces_joint_torque`, `test_identical_arms_give_identity_map`, `test_different_arms_over_render_joint_torque_reflection`.

The simulation shows the effect: reflecting the vertical force with the identity map over-renders it by a factor of 1.40, the Jacobian map by 1.10. Of the remaining excess, the position-channel term adds force when the slave sags under a sustained load.

Remark 15 (power). $\Gamma = J_m^\top S_f J_s^{+\top}$ is not power-preserving between the arms, because the arms are coupled in joint space ($q_m \leftrightarrow q_s$), not in task space. The slave-side power $\hat{\tau}^\top \dot{q}_s = \hat{F}^\top \dot{x}_s$ and the master-side power $\tau_f^\top \dot{q}_m = \hat{F}^\top S_f^\top \dot{x}_m$ are equal only if $\dot{x}_m = S_f^{-\top} \dot{x}_s$, which joint-space mapping does not enforce. This is why the force channel is wrapped by the energy tank.

6 Slave MPC on the position loop

6.1 What the draft's Eq. (7) said, and what was implemented

The draft's MPC minimises over torques $\tau_{0:N-1}$ of a rigid-body model and applies $\tau_0^\star$. The VX300s accepts position commands; no torque was ever sent to it. In the code the decision vector was $[dq_{0:N-1}, \tau_{0:N-1}]$ but dq never entered the rollout: the cost is $dq^\top R dq + \ldots$, so its minimiser is $dq = 0$ for any reference, and the value used, $dq_0^\star$, was identically zero. The slave moved only through the smoothing step $q \leftarrow 0.95q + 0.05q_{ref}$, a first-order lag of 0.39 s at 50 Hz. The joint-limit constraint of the draft was not enforced either (`q_min`, `q_max` were stored and never used). The revised formulation below replaces all of this.

6.2 Model and prediction

By Theorem 7 with $A = \begin{bmatrix} 0 & 1 \\ -\omega^2 & -2\zeta\omega \end{bmatrix}$, $B = \begin{bmatrix} 0 \\ \omega^2 \end{bmatrix}$, joint i of (2) gives $x[k+1] = A_d x[k] + B_d u[k]$ (the disturbance c is handled in Section 6.5). For one joint, with horizon N and model state $\hat{x}_0$, the predicted positions and velocities are

$$\hat{q}_k = e_1^\top A_d^k \hat{x}_0 + \sum_{j=0}^{k-1} e_1^\top A_d^{k-1-j} B_d \, u_j + \hat{d}, \qquad \hat{v}_k = e_2^\top A_d^k \hat{x}_0 + \sum_{j=0}^{k-1} e_2^\top A_d^{k-1-j} B_d \, u_j, \qquad k = 1, \ldots, N, \tag{10}$$

that is $\hat{q} = \Phi_q \hat{x}_0 + G_q U + \hat{d}\mathbf{1}$ and $\hat{v} = \Phi_v \hat{x}_0 + G_v U$ with $U = [u_0, \ldots, u_{N-1}]^\top$ and G_q, G_v lower-triangular Toeplitz.

6.3 The quadratic program

Let r_k be the reference preview ($r_k = q_{s,\mathrm{ref}} + k\Delta t \, S\dot{q}_m$), $w = (Q, \ldots, Q, P)$, $W = \mathrm{diag}(w)$, D the unit lower-bidiagonal difference matrix ($(DU)_k = u_k - u_{k-1}$) and $d_0 = [u_{-1}, 0, \ldots, 0]^\top$ with u_{-1} the last applied command:

$$\min_U \;\; (\hat{q} - r)^\top W (\hat{q} - r) + R \, \|DU - d_0\|^2 \quad \text{s.t.} \quad \begin{aligned} & q_{\min} \le u_k \le q_{\max}, \\ & |u_k - u_{k-1}| \le \Delta u_{\max}, \\ & |\hat{v}_k| \le \dot{q}_{\max}. \end{aligned} \tag{11}$$

Proposition 16 (condensed form)**.** *Let* $a = \Phi_q \hat{x}_0 + \hat{d}\mathbf{1} - r$*. Then* (11) *equals* $\frac{1}{2}U^\top H U + f^\top U + \text{const}$ *subject to* $l \le CU \le h$*, with*

$$H = 2\big(G_q^\top W G_q + R\, D^\top D\big), \quad f = 2G_q^\top W a - 2R\, D^\top d_0, \quad C = \begin{bmatrix} I \\ D \\ G_v \end{bmatrix}, \tag{12}$$

$l = [q_{\min}\mathbf{1};\ d_0 - \Delta u_{\max}\mathbf{1};\ -\dot{q}_{\max}\mathbf{1} - \Phi_v \hat{x}_0]$ *and* h *analogously.* $H \succ 0$ *for* $R > 0$*, so a feasible problem has a unique minimiser.*

Proof. Expand $(G_q U + a)^\top W (G_q U + a) + R(DU - d_0)^\top (DU - d_0)$ and collect the terms quadratic and linear in U. D is unit lower bidiagonal, hence invertible, so $D^\top D \succ 0$; $G_q^\top W G_q \succeq 0$ since $W \succeq 0$. The bounds follow by writing the three constraint groups in the variable CU. □

6.4 Solving it in under 2 ms: ADMM

With the auxiliary variable $Z = CU$, ADMM with proximal weight $\sigma > 0$ and penalty $\rho > 0$ iterates

$$U^+ = (H + \sigma I + \rho C^\top C)^{-1}\big(\sigma U - f + C^\top(\rho Z - y)\big), \tag{13}$$
$$Z^+ = \Pi_{[l,h]}\big(CU^+ + y/\rho\big), \qquad y^+ = y + \rho\,(CU^+ - Z^+), \tag{14}$$

(the OSQP scheme without relaxation [4], a special case of ADMM [5]).

Proposition 17 (ADMM)**.** *(i)* (13) *is the minimiser of the augmented Lagrangian* $\frac{1}{2}U^\top H U + f^\top U + \frac{\sigma}{2}\|U - U^k\|^2 + \frac{\rho}{2}\|CU - Z^k + y^k/\rho\|^2$*. (ii) A fixed point* (U, Z, y) *of* (13)–(14) *satisfies the KKT conditions of* (11)*:* $HU + f + C^\top y = 0$*,* $Z = CU \in [l, h]$*,* $y \in N_{[l,h]}(Z)$*. (iii) The matrix* $H + \sigma I + \rho C^\top C$ *does not depend on* $\hat{x}_0, r, u_{-1}, \hat{d}$ *or the bounds, so it is inverted once; an iteration costs one* $N \times N$ *product and one clipping. (iv) For any* $\rho, \sigma > 0$ *and a feasible convex QP the iterates converge to a KKT point [5, 4].*

Proof. (i) The gradient condition is $HU + f + \sigma(U - U^k) + \rho C^\top(CU - Z^k + y^k/\rho) = 0$, i.e. (13). (ii) At a fixed point the y-update gives $CU = Z$; the Z-update gives $Z = \Pi(Z + y/\rho)$, i.e. y/ρ in the normal cone of $[l, h]$ at Z; and (13) with $U^+ = U^k = U$, $Z^k = Z$ reduces to $HU + \rho C^\top CU = -f + \rho C^\top Z - C^\top y$, which with $CU = Z$ is $HU + f + C^\top y = 0$. (iii) is read off. (iv) is the cited result. □

Choice of ρ (a correction). A fixed number of iterations only approximates the optimum, and the approximation depends strongly on ρ. With $\rho = 10$ (the first implementation) and 60 iterations, the first move was up to 26 mrad from the exact solution from a cold start, and on average 13 mrad (maximum 48 mrad) when warm-started in closed loop: the controller was effectively a different, less regularised design than the documented QP. With $\rho = 0.3$ the warm-started first move is within 0.03 mrad on average (0.2 mrad maximum) of the exact solution, at the same cost (≈ 0.9 ms for five joints). The experiments in the repository were re-run with $\rho = 0.3$.

Numerical check: `test_admm_solves_the_stated_qp (reference: SLSQP on the same QP data), test_applied_command_always_respects_limits`.

Proposition 18 (the applied command is always admissible)**.** *After the solve,* $u_0^\star$ *is clipped to* $[q_{\min}, q_{\max}]$ *and to* $[u_{-1} - \Delta u_{\max}, u_{-1} + \Delta u_{\max}]$*. Since* u_{-1} *satisfied the same limits, the intersection is non-empty, and the applied command satisfies the position and rate limits regardless of solver accuracy (the predicted-velocity bound is enforced by the QP only).*

6.5 Offset-free tracking

The Dynamixel position loop has no gravity feed-forward, so under constant load joint i settles at $u + c/\omega^2$ rather than u. The model (10) cannot predict this. The controller runs the model in parallel with the plant (*internal-model* form): $\hat{x}[k+1] = A_d\hat{x}[k] + B_d u[k]$, and estimates an output disturbance

$$\hat{d}[k+1] = (1-\alpha_d)\hat{d}[k] + \alpha_d\big(q[k] - \hat{x}_1[k]\big), \qquad 0 < \alpha_d \le 1. \tag{15}$$

Proposition 19 (offset-free)**.** *Let the plant be (2) with constant c and let $u \equiv u_\infty$. Then $\hat{d} \to c/\omega^2$. Moreover, if the reference is constant, $r \equiv r_\infty$, and $u_\infty = r_\infty - \hat{d}$ lies within the limits, then $(u_\infty,\ \hat{d} = c/\omega^2)$ is an equilibrium of the closed loop at which the plant position equals r_∞.*

Proof. The plant converges to $q_\infty = u_\infty + c/\omega^2$ (A_d is Schur by Theorem 7(i)); the parallel model converges to $\hat{x}_{1,\infty} = u_\infty$. Hence $q - \hat{x}_1 \to c/\omega^2$ and the stable recursion (15) (pole $1-\alpha_d$) converges to the same value. For the second claim, with $\hat{d} = c/\omega^2$ and $U \equiv u_\infty = r_\infty - \hat{d}$ the predictions satisfy $\hat{q}_k + \hat{d} = r_\infty$ for all k and $\Delta u = 0$, so the cost of (11) is 0, its global minimum; by Proposition 16 the minimiser is unique, so the MPC outputs u_∞, and the plant sits at $u_\infty + c/\omega^2 = r_\infty$. □

Numerical check: `test_offset_free_estimate_converges_and_removes_steady_state_error` (sag 48 `mrad removed;` $\hat{d}$ `within` 2%).

This is the standard offset-free MPC mechanism [6, 7]. Proposition 19 establishes the existence of the zero-error equilibrium, not its asymptotic stability; the latter is checked in simulation only.

6.6 What is not claimed

No terminal cost or terminal set with a Lyapunov argument is used, so *closed-loop stability of the MPC is not proven*; $N = 10$ and the weights were chosen by simulation. The prediction model is decoupled per joint and validated in simulation only (Section 10).

7 Energy tank on the force channel

Let $P_{out}[k] = \tau_f[k]^\top \dot{q}_m[k]$ be the power the (unscaled) force channel delivers to the operator–master system, and $P_{in}[k] = \eta\big(\dot{q}_m^\top D_{md}\dot{q}_m + \dot{q}_e^\top D_{sd}\dot{q}_e\big)[k] \ge 0$ ($0 < \eta \le 1$) the power the controller dissipates through its two damping terms. The tank stores $T[k] \in [T_{\min}, T_{\max}]$, $0 < T_{\min} \le T[0] \le T_{\max}$, and a gate $\gamma[k] \in [0,1]$ scales the force:

$$\gamma[k] = \begin{cases} 1 & \text{if } P_{out}[k] \le 0 \ \text{ or } \ b[k] \ge \Delta t\, P_{out}[k], \\ \dfrac{b[k]}{\Delta t\, P_{out}[k]} & \text{otherwise,} \end{cases} \qquad b[k] = T[k] + \Delta t P_{in}[k] - T_{\min}, \tag{16}$$

$$T[k+1] = \min\big\{T_{\max},\ T[k] + \Delta t\,(P_{in}[k] - \gamma[k]P_{out}[k])\big\}. \tag{17}$$

Theorem 20 (bounded delivered energy)**.** *For any signals $\tau_f, \dot{q}_m, P_{in} \ge 0$ and every $K \ge 1$: (i) $\gamma[k] \in [0,1]$ and $T[k] \ge T_{\min}$ for all k; (ii) the energy delivered by the gated force channel is bounded:*

$$\sum_{k=0}^{K-1} \Delta t\, \gamma[k] P_{out}[k] \ \le\ T[0] - T_{\min} + \sum_{k=0}^{K-1} \Delta t\, P_{in}[k]. \tag{18}$$

Proof. (i) Induction on $T[k] \ge T_{\min}$. Since $P_{in} \ge 0$, $b[k] \ge 0$, so $\gamma \ge 0$; and $\gamma \le 1$ by construction. If $P_{out} \le 0$, $\gamma = 1$ and $T[k+1] = \min\{T_{\max}, T[k] + \Delta t(P_{in} + |P_{out}|)\} \ge \min\{T_{\max}, T[k]\} \ge T_{\min}$. If $P_{out} > 0$ and $b \ge \Delta t P_{out}$, then $\gamma = 1$ and $T[k] + \Delta t(P_{in} - P_{out}) = b + T_{\min} - \Delta t P_{out} \ge T_{\min}$. Otherwise $\gamma \Delta t P_{out} = b$ and $T[k] + \Delta t P_{in} - b = T_{\min}$. In every case the argument of the min in (17) is $\ge T_{\min}$ and $T_{\max} \ge T_{\min}$. (ii) Write (17) as $T[k+1] = T[k] + \Delta t(P_{in} - \gamma P_{out}) - c[k]$ with the clipping loss $c[k] \ge 0$. Summing over k, $\sum \Delta t \gamma P_{out} = T[0] - T[K] + \sum \Delta t P_{in} - \sum c[k] \le T[0] - T_{\min} + \sum \Delta t P_{in}$, using $T[K] \ge T_{\min}$ and $c \ge 0$. □

Numerical check: `test_tank_energy_bound_theorem (random signals, 5000 steps)`, `test_energy_tank_never_below_minimum`.

Reading of the result. Whatever the force estimate does, the force channel can return to the operator at most the energy that the controller has dissipated plus the initial reserve $T[0] - T_{\min}$; $T_{\max}$ caps how much can be released in a burst. Combined with Proposition 12, the master with the gated feedback has $\dot{V} \le \dot{q}^\top \tau_h - \dot{q}^\top D \dot{q} + \gamma P_{out}$, with the last term energetically bounded by (18).

What it does not say.

- (18) uses the Riemann sum $\Delta t\, \tau_f^\top \dot{q}_m$ for the energy of a sample; it is exact only to the discretisation error of that product. An exact zero-order-hold energy would use $\tau_f[k]^\top (q_m[k+1] - q_m[k])$, which needs the next sample.
- It bounds the energy delivered by the *master-side controller*. With communication delay the slave-side quantities in τ_f are stale, and a delayed channel is not passive in general [10]; the tank does not address this. The simulations confirm it: the tank enlarges the stable force gain without delay (from $k_{of} \approx 0.6$ to ≈ 1.0–1.5) and gives no benefit at 20–60 ms of delay. A two-port tank or time-domain passivity control with energy exchange between the sites [8, 9] is needed for delayed operation.
- Passivity of the controller port together with a passive operator and environment yields stability of the interconnection by the passivity theorem; neither passivity is verified here.

8 Metrics

For errors $e[k, j]$ over K samples and J joints, $\mathrm{RMSE}_j = \sqrt{\frac{1}{K}\sum_k e[k,j]^2}$ and the *pooled* RMSE is $\mathrm{RMSE}_{\mathrm{pool}} = \sqrt{\frac{1}{KJ}\sum_{k,j} e[k,j]^2} = \sqrt{\frac{1}{J}\sum_j \mathrm{RMSE}_j^2}$.

Proposition 21. $\frac{1}{J}\sum_j \mathrm{RMSE}_j \le \mathrm{RMSE}_{\mathrm{pool}}$, *with equality iff all* RMSE_j *are equal.*

Proof. By Jensen's inequality (convexity of $x \mapsto x^2$), $\left(\frac{1}{J}\sum_j a_j\right)^2 \le \frac{1}{J}\sum_j a_j^2$ for $a_j = \mathrm{RMSE}_j$; equality iff all a_j coincide. □

For the draft's Table I the per-joint values $0.2250, 0.2715, 0.3609, 0.3152, 0.5366$ give a mean of 0.3418 and a pooled value of 0.3583 rad, so the table's "Overall" row is the pooled one and the 0.3418 quoted in the conclusion was the mean of the joint RMSEs.

Numerical check: `test_mean_of_joint_rmse_never_exceeds_pooled_rmse`, `test_pooled_rmse_is_not_mean_of_rmse`.

The *lag* of joint j is $s^\star \Delta t$ with $s^\star = \arg\max_{0 \le s \le s_{\max}} \rho(q_m[0{:}K{-}s],\, q_s[s{:}K])$ (ρ the Pearson correlation), at a resolution of one sample (20 ms); the *lag-compensated RMSE* is the RMSE of $q_s[s^\star{:}] - q_m[{:}K{-}s^\star]$. Definitions of the force metrics are in `docs/EVALUATION_METHODOLOGY.md`.

9 From the draft's equations to the revised ones

Draft	Issue	Revised
Eq. (1)	$\tau_m = \mathcal{F}\{k_t \odot I\}$ with I in mA treated as A (1000× too large); two-motor joints counted once; nominal Δt in the finite difference	$\tau_m = n_\mu \odot k_t \odot 10^{-3} I$ (Assumption 2); message time stamps in the velocity
Eq. (2)	$\tau_m - (C - G)$: sign of G, C used as a vector, $M\ddot{q}$ ignored (Lemma 1)	slave: momentum observer, Theorem 3; master: no estimate (it only echoes its own command)
Eq. (3),(4)	$\tau_{m,cmd}$ contains $\hat{\tau}_{ext,m}$: integrates the command, Lemma 11	Eq. (7)
Eq. (5)	Euler step with the physical inertia as M_{sd} and $\varepsilon = 10^{-4}$	design M_d, D_d, K_d with the exact discretisation (Theorem 7); the operator-force term is removed with the master estimate
Eq. (6)	joint-by-joint reflection; called four-channel	Jacobian map (9) and energy tank; with the master estimate removed the scheme has *three* channels
Eq. (7)	torque MPC applied as $\tau_0^\star$; the code optimised a dead variable and ignored q limits	position-command QP (11), offset-free, applied as $u_0^\star$

10 What is not proven, and what hardware must confirm

1. **Closed-loop stability** of the MPC (no terminal ingredients) and of the full teleoperation loop, in particular with delay.
2. **Model validity on the real arm**: decoupled second-order position loop with a constant disturbance (Assumption 3); ω_i, ζ_i in the repository were identified on the *simulated* slave. Backlash, the Dynamixel velocity/acceleration profile and current limiting are not modelled.
3. **Torque constants and inertial parameters** (Remark 4): the observer's accuracy is limited by k_t, link masses and friction, which must be calibrated.
4. **Energy accounting** is sampled-data (Section 7); the damping-dissipation term P_{in} assumes the master's and slave's damping terms are implemented as written.
5. The **dead zones, gains and** N were chosen in simulation.

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